Optimization of Structural Material Mix

Multi-period Modeling for Low-Carbon Construction

HAJLAOUI Khaoula and ZOUARHI Yassmin
Under the supervision of Mrs.CHLELA Sophie

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1 Introduction

1.1 Context

The building and construction sector is by far the largest emitter of greenhouse gases, accounting for 37 % of global emissions. The production and use of materials such as cement, steel, and aluminum have a significant carbon footprint.

Historically, most progress in the sector has focused on reducing *operational carbon emissions* of buildings, namely emissions resulting from heating, cooling, and lighting. However, as operational emissions decrease due to improvements in energy efficiency, the relative importance of emissions associated with the construction phase and material production — often referred to as *embodied emissions* — becomes increasingly significant.

In this context, the present project shifts the focus from operational performance to the optimization of construction-related impacts by determining an optimal mix of structural materials that meets climate objectives while respecting economic and land-use constraints.

1.2 Project Objectives

The main objective of this project is to develop an optimization framework for planning low-carbon construction strategies over a multi-period horizon. More specifically, the project aims to:

- Develop a multi-period linear optimization model describing the allocation of structural materials over time;
- Minimize the total system cost while accounting for carbon emissions through a carbon pricing mechanism;
- Integrate the role of carbon sinks associated with bio-based materials and afforestation dynamics;
- Incorporate land-use constraints linked to forest resources and ecosystem protection policies;
- Evaluate the impact of carbon taxes and environmental policies on material selection;
- Compare different policy scenarios, including a reference case and a land-protection (30×30) scenario.

1.3 Problem Description

This project focuses on modeling how material allocation decisions can be optimized under environmental and economic constraints.

1.3.1 Structural Materials

Three structural systems are considered in this study, each characterized by different economic costs, carbon intensities, and resource implications.

Concrete is currently the most widely used construction material due to its low cost and mechanical robustness. However, cement production is highly energy-intensive and represents a major source of global CO_2 emissions.

Steel offers high structural performance and recyclability but requires energy-intensive industrial processes, resulting in significant carbon emissions despite technological improvements.

Cross-Laminated Timber (CLT) is a bio-based material capable of storing carbon during its lifecycle. Its use can reduce embodied emissions, but large-scale deployment is constrained by forest availability and land-use considerations.

1.3.2 Time Horizon: 2025–2050

The analysis is conducted over the 2025–2050 period, which corresponds to the timeframe commonly adopted in international climate strategies aiming for carbon neutrality by mid-century. The planning horizon is discretized into time periods of five years, allowing the model to capture medium-term transitions while keeping the optimization problem computationally tractable. This long-term perspective enables the evaluation of transition pathways and the progressive adaptation of material production capacities and environmental policies.

1.3.3 Policy Background

Environmental policies increasingly combine carbon pricing mechanisms with land-protection strategies to balance climate mitigation and biodiversity preservation. Carbon taxes aim to internalize environmental costs by penalizing high-emission materials, while land-use regulations limit the exploitation of forest resources and encourage sustainable afforestation.

1.3.4 Studied Scenarios

Two scenarios are analyzed in order to evaluate the impact of environmental policies on optimal material allocation:

- **Reference scenario:** no additional land-use restrictions are imposed, representing a baseline trajectory driven mainly by economic costs.
- **30×30 policy scenario:** at least 30 % of total land is protected starting in 2030, reflecting international biodiversity targets. This constraint limits forest exploitation and directly affects the availability of timber-based construction materials.

2 Mathematical Modeling

2.1 Optimization Framework

This project is formulated as a linear optimization problem. Optimization techniques are widely used in engineering, in particular, optimization models are commonly applied to energy transition and climate policy studies in order to identify cost-efficient strategies that satisfy sustainability targets.

In this work, optimization is used to determine the optimal allocation of construction materials and land resources over time while minimizing total system cost and accounting for carbon emissions and policy constraints.

2.2 Decision Variables

Decision variables represent the quantities whose values are determined by the optimization model. They correspond to the choices made by the model in order to achieve the objective while satisfying all constraints.

The main decision variables considered in the model are:

- $x_{conc,t}$: built surface area using concrete in period t [m^2]
- $x_{steel,t}$: built surface area using steel in period t [m^2]
- $x_{clt,t}$: built surface area using Cross-Laminated Timber (CLT) in period t [m^2]
- $cap_{clt,t}$: available production capacity of CLT in period t [Mt]

- $a_{timber,t}$: forest area allocated for timber production in period t [kha]
- $a_{aff,t}$: afforestation area in period t [kha]
- $a_{protected,t}$: protected land area in period t [kha]
- $n_{timber,t}$: net variation of timber area in period t [kha]

Derived variables. Some quantities are not directly optimized but are computed from decision variables through fixed coefficients:

- $q_{cem,t}$: cement quantity derived from concrete surface;
- $q_{steel,t}$: steel quantity derived from steel structures;
- $q_{clt,t}$: CLT quantity derived from timber construction.

Model parameters. Other quantities are exogenous inputs provided to the model, such as material intensities, demand projections, emission factors, costs, and total land availability. These parameters are fixed and are not determined by the optimization.

2.3 Objective Function

The objective is to minimize the total system cost over the planning horizon, including construction costs, afforestation costs, and carbon costs:

$$\min \sum_t \left[C_{cem}\, q_{cem,t} + C_{steel}\, q_{steel,t} + C_{clt}\, q_{clt,t} + C_{aff}\, a_{aff,t} + \tau_t \Big(E_{mat,t} + LUC_t - M_{aff,t} - M_{clt,t} \Big) \right]$$

where:

- C_i: unit cost of material i [€/t]
- C_{aff}: afforestation cost [€/ha]
- τ_t: carbon price in year t [€/tCO_2]
- $E_{mat,t}$: emissions from material production in year t [tCO_2]
- LUC_t: emissions from land-use change in year t [tCO_2]
- $M_{aff,t}$: carbon sequestration from afforestation in year t [tCO_2]
- $M_{clt,t}$: carbon credit from CLT in year t [tCO_2]

2.4 Constraints

2.4.1 Demand Satisfaction

The total constructed area must satisfy the projected demand S_t:

$$x_{conc,t} + x_{steel,t} + x_{clt,t} \geq S_t, \quad \forall t$$

2.4.2 Material Quantities

The quantities of cement, steel, and CLT are determined by the built surfaces and material intensities:

$$
\begin{aligned}
q_{cem,t} &= MI_{cem}^{conc} x_{conc,t} + MI_{cem}^{steel} x_{steel,t} + MI_{cem}^{clt} x_{clt,t} \\
q_{steel,t} &= MI_{steel}^{conc} x_{conc,t} + MI_{steel}^{steel} x_{steel,t} + MI_{steel}^{clt} x_{clt,t} \\
q_{clt,t} &= MI_{clt}^{clt}\, x_{clt,t}
\end{aligned}
$$

2.4.3 Land-Use Constraints

The total land used for timber, afforestation, and protected areas cannot exceed the total available land:

$$a_{timber,t} + a_{aff,t} + a_{protected,t} \leq A_{tot}, \quad \forall t$$

2.4.4 CLT Production Capacity and Ramping

The quantity of CLT cannot exceed the production capacity:

$$q_{clt,t} \leq cap_{clt,t} \cdot 10^6, \quad \forall t$$

The production capacity is ramped over time:

$$cap_{clt,t} \leq cap_{clt,t-1} + \Delta cap, \quad \forall t > t_0$$

Similarly, the expansion of timber area is limited by a maximum annual increase:

$$a_{timber,t} \leq a_{timber,t-1} + \Delta a, \quad \forall t > t_0$$

2.4.5 Technical and Adoption Constraints

- Minimum cement for foundations:

$$q_{cem,t} \geq \gamma_{fond} \left(x_{conc,t} + x_{steel,t} + x_{clt,t} \right)$$

- CLT adoption ramping:

$$|x_{clt,t} - x_{clt,t-1}| \leq adopt_ramp$$

- Maximum technical eligibility of CLT:

$$x_{clt,t} \leq \beta_{CLT}\, S_t$$

- Minimum share of concrete:

$$x_{conc,t} \geq \alpha_{conc}\, S_t$$

- Progressive minimum CLT adoption:

$$x_{clt,t} \geq \text{CLT_target}_t\, S_t$$

2.4.6 30×30 Policy Constraint

Under the 30×30 policy, at least 30% of land must be protected from 2030 onward:

$$a_{protected,t} \geq 0.3\, A_{tot}, \quad \forall t \geq 2030$$

2.4.7 Net Zero Land-Use Change (LUC)

The net carbon impact from land-use change, afforestation, and CLT carbon storage must be zero over the planning horizon:

$$\sum_t (LUC_t - M_{aff,t} - M_{clt,t}) = 0$$

2.4.8 Model Adjustment and CLT Adoption Constraints

Initial simulations of the optimization model resulted in solutions where CLT was almost entirely avoided in favor of conventional materials. Although economically optimal under the initial formulation, such outcomes were not consistent with current policy objectives and observed technological transition pathways promoting bio-based construction materials.

To better represent realistic adoption dynamics, additional constraints were introduced to enforce a progressive deployment of CLT over time. In particular, minimum adoption targets were imposed, requiring CLT construction to represent a given share of total building demand at each period.

3 Model Assumptions

The optimization model is based on the following simplifying assumptions:

- Linear relationships are assumed between built surfaces, material quantities, costs, and emissions.
- Carbon emissions and carbon sequestration effects are additive across materials and land-use activities.
- CLT production depends on available forest resources and evolves progressively due to capacity and adoption ramping constraints.
- Land allocation between timber production, afforestation, and protected areas is mutually exclusive and limited by total available land.
- Construction demand is deterministic and known in advance for each period.
- Carbon pricing and policy parameters are exogenous and fixed over the planning horizon.
- The planning horizon is discretized into five-year periods (2025–2050).

4 Results

This section presents the optimization outputs obtained for the two studied scenarios. The results are analyzed in terms of material allocation, land-use dynamics, economic costs, and carbon emissions in order to evaluate the impact of environmental policies on the construction system.

4.1 Reference Scenario

The reference scenario represents the baseline configuration of the model, in which construction decisions are driven by economic costs and carbon pricing mechanisms without biodiversity protection constraints.

The optimization determines the optimal allocation of construction surfaces, land use, and material quantities over time while satisfying technical and environmental constraints.

Figure 1 shows the distribution of construction materials in 2050. The results indicate a dominance of conventional materials, with steel representing approximately 80% of constructed surfaces and concrete 20%. No CLT deployment occurs in this scenario, as its higher direct production cost prevents adoption despite carbon pricing incentives.

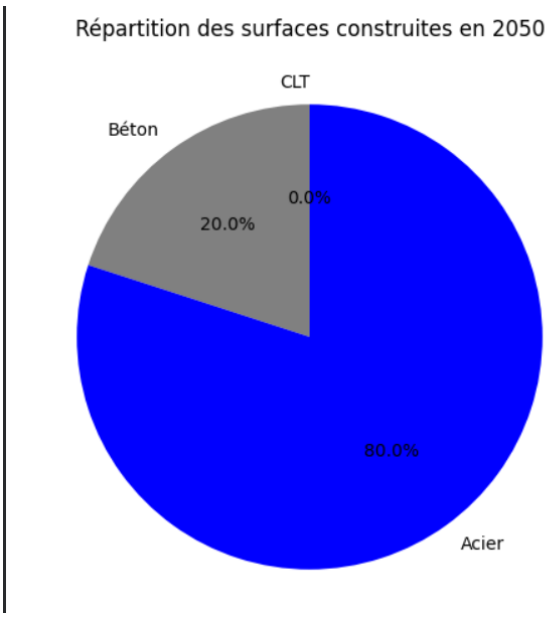

Figure 1: Distribution of construction materials in 2050 – Reference scenario

From an environmental perspective, Figure 2 illustrates that emissions remain positive throughout the planning horizon. Since neither afforestation nor wood-based construction is adopted, no additional carbon sequestration is generated, preventing the system from achieving negative emissions.

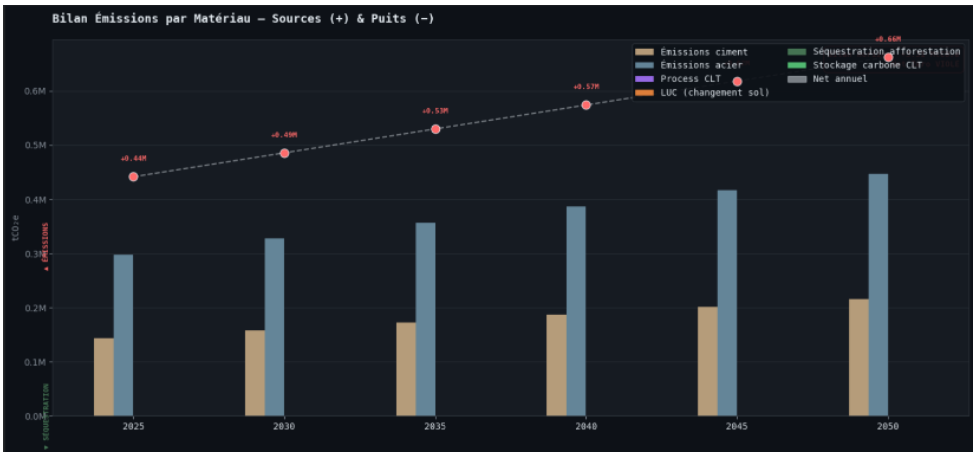

Figure 2: Carbon emissions over time – Reference scenario

The cost structure shown in Figure 3 is largely dominated by steel (78.5%) and concrete (21.5%), reflecting the absence of bio-based construction pathways.

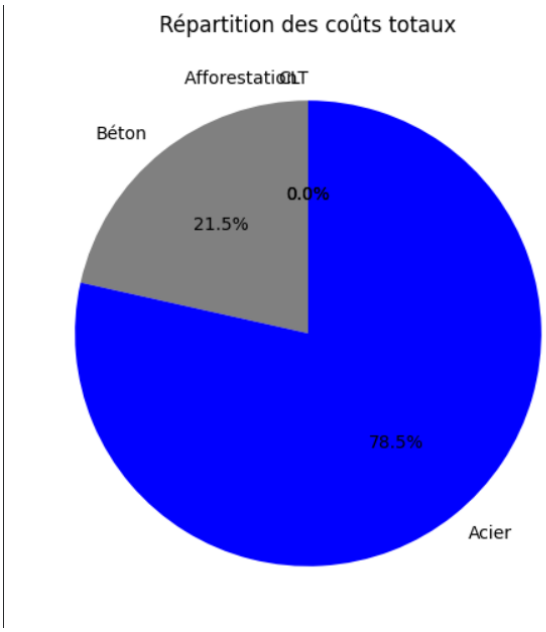

Figure 3: Cost distribution – Reference scenario

4.2 30×30 Policy Scenario

The 30×30 scenario introduces a biodiversity protection constraint requiring at least 30 % of total land to be protected from 2030 onward. This assumption is aligned with the *EU Biodiversity Strategy for 2030* and Target 3 of the *Kunming–Montréal Global Biodiversity Framework*, which aim to preserve 30 % of terrestrial ecosystems.

All remaining parameters are identical to the reference scenario, allowing a direct evaluation of policy impacts.

Figure 4 shows a major transition in the construction system. By 2050, CLT represents around 60% of constructed surfaces, while steel and concrete each account for roughly 20%. This shift results from the interaction between carbon pricing, carbon storage credits, and land-based climate strategies.

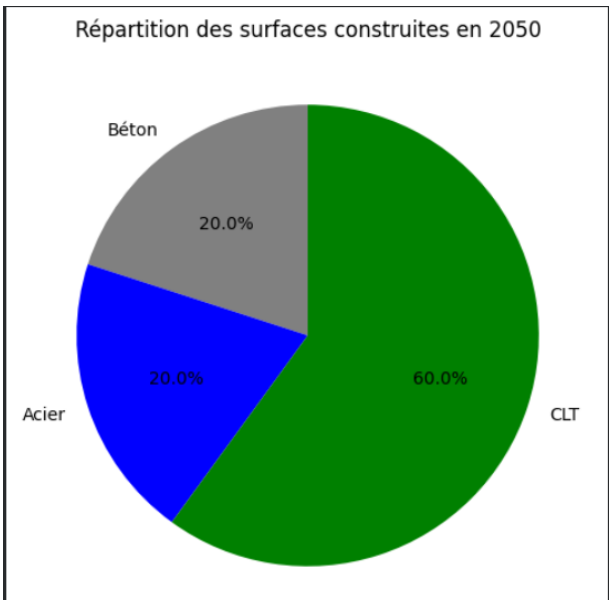

Figure 4: Distribution of construction materials in 2050 – 30×30 scenario

The emissions trajectory presented in Figure 5 reveals a structural change compared to the reference case. While early periods exhibit positive emissions due to land-use change, the system achieves net-negative emissions from 2035 onward. This outcome is driven by afforestation sequestration and long-term carbon storage in timber materials.

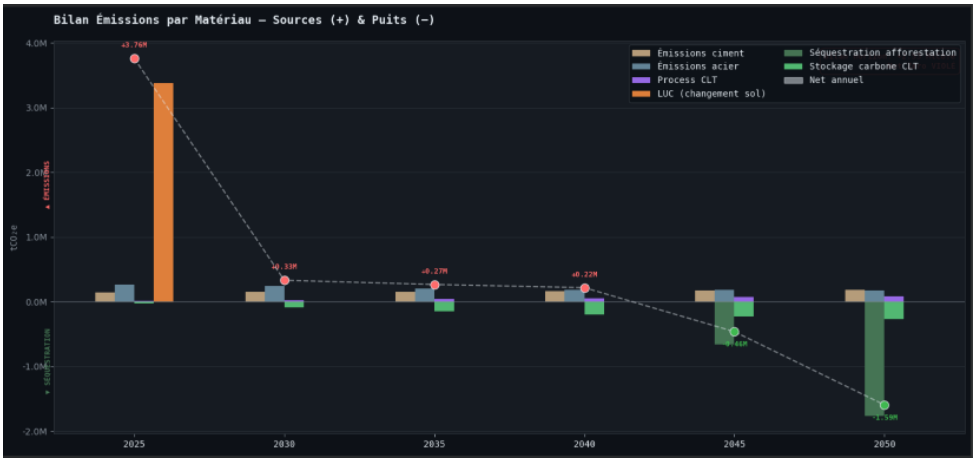

Figure 5: Carbon emissions over time – 30×30 scenario

Figure 6 shows a more diversified cost structure, with CLT representing the largest share (61%), followed by steel (17%), afforestation investments (12.1%), and concrete (9.9%).

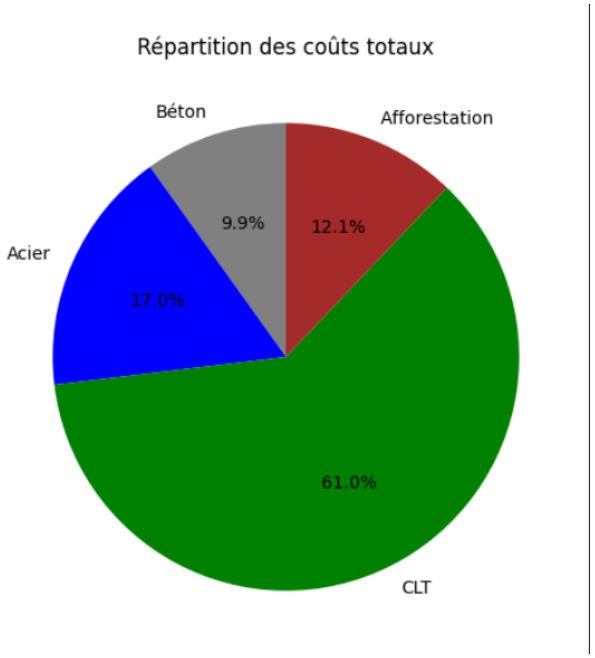

Figure 6: Cost distribution – 30×30 scenario

4.3 Comparative Analysis

The comparison between scenarios highlights key economic and environmental trade-offs.

First, carbon pricing plays a central role in driving material substitution. As carbon prices increase over time, cement and steel become relatively more expensive due to their higher emission factors. Although CLT involves higher direct production costs, its lower lifecycle emissions and associated carbon storage credits make it economically competitive under stricter environmental conditions.

Second, biodiversity protection reshapes land-use allocation and indirectly influences construction choices. While the reference scenario maintains reliance on conventional materials, the 30×30 policy promotes the integration of nature-based solutions, enabling large-scale deployment of timber construction.

Finally, the results demonstrate a strong synergy between land policy and climate mitigation. The 30×30 scenario simultaneously supports biodiversity preservation, carbon sequestration through afforestation, and decarbonization of the construction sector, ultimately achieving net-negative emissions over the planning horizon.

5 Shadow Price Analysis (Dual)

Shadow prices (dual values) measure the sensitivity of the objective function to each constraint: a dual value of λ_i means that relaxing the constraint by one unit reduces the total cost by λ_i euros.
In a linear program of the form:

$$\min_x \; c^T x \quad \text{s.t.} \quad Ax \geq b, \quad x \geq 0 \tag{1}$$

The dual variable λ_i associated with constraint i is obtained by solving the dual problem:

$$\max_\lambda \; b^T \lambda \quad \text{s.t.} \quad A^T \lambda \leq c, \quad \lambda \geq 0 \tag{2}$$

In practice, PuLP retrieves the dual values directly from the CBC solver after resolution via `constraint.pi`.

5.1 Net Zero LUC Constraint

The shadow price of the net zero LUC constraint is 60 €/tCO_2, meaning that allowing one additional tonne of CO_2 emissions would reduce total cost by 60 €. This represents the implicit carbon price of the system.

5.2 Progressive CLT Minimum

The dual value of 88 €/m^2 in 2035 indicates that CLT is more expensive than the optimal mix at that period. CLT adoption is therefore policy-driven, not cost-optimal.

5.3 Forest Production (2050)

The shadow price of −275 €/tonne in 2050 shows that timber is a critical and binding resource. Expanding forest production capacity would significantly reduce total system cost.

5.4 Land Use (2050)

The dual value of −100,000 €/kha in 2050 reveals that land is the most binding constraint in the final period. This is the strongest signal in the model: land scarcity is the primary bottleneck of the transition.

6 Discussion

6.1 Sensitivity of the Optimal Solution

The dual analysis reveals three binding constraints that most strongly drive the total cost. Land scarcity in 2050 presents the most critical bottleneck, with a dual value of −100,000 €/kha,

meaning that 1 kha of additional land would reduce total cost by 100,000 €. The Net Zero LUC constraint yields an implicit carbon price of 60 €/tCO_2, confirming that carbon neutrality comes at a significant economic cost. Finally, the marginal cost of demand rises from 30.92 to 39.44 €/m² between 2025 and 2050 (+27%), reflecting the cumulative pressure of carbon and land constraints over time.

6.2 Limitations

The model assumes perfect information over all periods (2025–2050), which is unrealistic in practice. It is also deterministic and does not account for price volatility, supply chain disruptions, or technological uncertainty. Furthermore, the model operates at a global aggregated scale, without capturing regional variations in forest productivity, construction practices, or transportation costs. This may mask local trade-offs and implementation challenges associated with land-use policies.

Regarding CLT, its dual values are consistently positive throughout the optimization horizon, meaning the solver only adopts it when forced by policy constraints. This suggests that without regulatory intervention, the market would not naturally transition to timber construction under current cost parameters.

6.3 Regional Perspectives on CLT Adoption

CLT's high cost in the model reflects current market conditions in Europe and most of the Global South. However, the picture differs significantly by region. Canada, one of the world's largest timber producers, already uses CLT extensively in regions such as British Columbia, where it can be cost-competitive with steel and concrete. In Central and West Africa, countries such as Cameroon, Gabon, and the Democratic Republic of Congo hold vast tropical timber reserves where, with appropriate investment, CLT production could be both economically viable and a driver of local industrialization. Nordic countries such as Sweden and Finland are already leaders in mass timber construction, where CLT approaches cost parity due to strong forest industries and carbon pricing policies.

6.4 Role of Industry in Accelerating CLT Adoption

The ramping constraints in the model reflect real-world inertia in CLT adoption. Key industry levers to overcome this include scaling up manufacturing capacity to reduce unit costs, standardizing CLT building codes to lower design and certification costs, investing in supply chains linking forest owners, sawmills, and construction firms, and enforcing public procurement policies mandating timber in public buildings, as already implemented in France (*loi RE2020*) and Austria.

6.5 Link with Climate Studies

The results of this model are consistent with findings from the climate literature. The IPCC AR6 report identifies the substitution of high-emission materials by bio-based alternatives as a key mitigation lever in the built environment, with a potential of up to 0.6 GtCO_2/year by 2050. Studies on carbon storage in harvested wood products confirm that CLT buildings act as long-term carbon sinks, consistent with the S_{CLT} credit parameter used in our model. Finally, the 30×30 biodiversity target directly constrains land availability in our model, and climate studies warn that failure to protect ecosystems will increase land-use emissions — exactly what the Net Zero LUC constraint is designed to prevent.

7 Conclusion

This study developed a multi-period linear optimization model to analyze the transition toward low-carbon construction systems under economic, environmental, and land-use constraints. The model integrates material allocation decisions, carbon pricing mechanisms, afforestation dynamics, and biodiversity protection policies within a unified optimization framework covering the 2025–2050 horizon.

The results highlight the central role of policy design in shaping the evolution of construction systems. In the reference scenario, where decisions are primarily driven by economic costs, the optimization favours conventional materials such as steel and concrete. Despite the presence of carbon pricing, bio-based materials remain limited due to their higher direct costs and resource constraints, resulting in persistent positive emissions over time.

In contrast, the introduction of the 30×30 biodiversity policy produces a structural transformation of the system. The protection of land, combined with carbon pricing incentives, promotes the deployment of Cross-Laminated Timber (CLT) and afforestation strategies. These nature-based solutions enable carbon sequestration and long-term carbon storage, allowing the system to achieve net-negative emissions while maintaining demand satisfaction.

The comparative analysis demonstrates that climate mitigation and biodiversity protection policies are not necessarily conflicting. Instead, their interaction can generate synergies that accelerate the transition toward sustainable construction pathways. However, the results also reveal important trade-offs, particularly increased system costs and stronger dependence on limited land resources.

Shadow price analysis further confirms that land availability and forest resources constitute the main bottlenecks of the transition, suggesting that future decarbonization strategies must carefully balance material innovation with sustainable land management.

Future research perspectives. Several extensions could improve the realism and policy relevance of the model. First, introducing uncertainty through stochastic or robust optimization could better capture future variability in carbon prices, technological costs, and demand projections. Second, incorporating regional spatial differentiation would allow analysis of local resource constraints and transportation effects. Third, nonlinear technological learning mechanisms could represent cost reductions associated with large-scale deployment of low-carbon materials. Finally, expanding the policy framework to include subsidies, recycling dynamics, or circular economy strategies would provide a more comprehensive assessment of decarbonization pathways in the construction sector.

Overall, this work demonstrates the usefulness of optimization-based approaches for supporting long-term decision-making at the intersection of climate policy, biodiversity conservation, and sustainable infrastructure planning.

References

[1] International Energy Agency (IEA). *Buildings Sector Emissions Report*. IEA Publications, 2023. https://www.iea.org/reports/buildings

[2] UNEP. *2022 Global Status Report for Buildings and Construction*. United Nations Environment Programme, 2022. https://globalabc.org/resources/publications/2022-global-status-report-buildings-and-construction

[3] IPCC. *Climate Change 2022: Mitigation of Climate Change*. Working Group III Contribution to the Sixth Assessment Report. Cambridge University Press, 2022. https://www.ipcc.ch/report/ar6/wg3/

[4] European Commission. *EU Biodiversity Strategy for 2030*. European Union, 2020. https://environment.ec.europa.eu/strategy/biodiversity-strategy-2030_en

[5] Convention on Biological Diversity. *Kunming–Montreal Global Biodiversity Framework*. United Nations, 2022. https://www.cbd.int/gbf/

[6] Churkina, G., Organschi, A., Reyer, C. et al. Buildings as a global carbon sink. *Nature Sustainability*, 3, 269–276 (2020). https://doi.org/10.1038/s41893-019-0462-4

[7] Habert, G., Miller, S. A., John, V. M., et al. Environmental impacts and decarbonization strategies in the cement and concrete industries. *Nature Reviews Earth & Environment*, 1, 559–573 (2020).

[8] Luderer, G. et al. Residual fossil CO_2 emissions in 1.5–2°C pathways. *Nature Climate Change*, 8, 626–633 (2018).

[9] Bertsimas, D., and Tsitsiklis, J. *Introduction to Linear Optimization*. Athena Scientific, 1997.